

非正态分布, Jackknife resample → 检验方法
 随机效应向量预测 → BLUP, LUP, AUP
 固定效应向量估计 → BLUE (GLS, OLS)

Chapter 10. Statistical Analysis of Mixed Linear Model

10.1 Mixed Linear Model

10.2 The Analysis of Random Effects in Mixed Linear Model

10.3 Estimation of Fixed Effects in Mixed Linear Model

10.4 Examples on the Application of Mixed Linear Model

Why use the analysis of mixed linear model (I)

Multiple Regression Model: $y = Xb + e \sim MVN(Xb, \sigma^2 I)$

- The independent variables are fixed effects, unable to analyze variables of random effects
- The independent variables are not independent of each other, the coefficient of regression is partial
- Only the random residual errors are independent of each other

ANOVA Model: $y_{ijk} = \mu + A_i + B_j + AB_{ij} + \varepsilon_{ijk}$ (大量重复/区组)

- The total variation can be decomposed into different variation components
- Unable to analyze complicated statistical model (the coefficients are not integers, the factors are not independent of each other, the factors are easily confused, and so on)
- Unable to analyze unbalanced data effectively
- Variances of random effects can be estimated, but can't for the values of random effects
- Lower power for analysis of parameters

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变异分解, 检验效应的显著性.

固定效应下的分解都是全局MSE. 混合/随机效应下的分解会因模型变化.

方差分析模型都可以写成线性模型的形式

区域试验数据:

3 varieties (fixed)

2 years (random)

2 locations (random)

2 blocks (random)

1	1	1	1	19.7	2	1	1	1	28.6	3	1	1	1	20.3
1	1	1	2	31.4	2	1	1	2	38.3	3	1	1	2	27.5
1	1	2	1	40.8	2	1	2	1	44.4	3	1	2	1	44.6
1	1	2	2	29.4	2	1	2	2	34.9	3	1	2	2	41.4
1	2	1	1	45.5	2	2	1	1	47.5	3	2	1	1	54.2
1	2	1	2	50.3	2	2	1	2	41.1	3	2	1	2	52.3
1	2	2	1	53.9	2	2	2	1	63.7	3	2	2	1	53.9
1	2	2	2	58.8	2	2	2	2	61.1	3	2	2	2	59.1

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左边第1-4列分别为品种、年份、地点、区组代码

$$Y_{hijk} = \mu + \alpha_h + \beta_i + \gamma_j + \alpha\beta_{hi} + \alpha\gamma_{hj} + \beta\gamma_{ij} + \varepsilon_{hijk}$$

$h = 1, 2, 3; i, j, k = 1, 2$

1	1	1	1	19.7	1	1	1	1	28.6	1	1	1	1	20.3
1	1	1	2	31.4	1	1	1	2	38.3	1	1	1	2	27.5
1	1	2	1	40.8	1	1	2	1	44.4	1	1	2	1	44.6
1	1	2	2	29.4	1	1	2	2	34.9	1	1	2	2	41.4
1	2	1	1	45.5	1	2	1	1	47.5	1	2	1	1	54.2
1	2	1	2	50.3	1	2	1	2	41.1	1	2	1	2	52.3
1	2	2	1	53.9	1	2	2	1	63.7	1	2	2	1	53.9
1	2	2	2	58.8	1	2	2	2	61.1	1	2	2	2	59.1

品种 年份 地点

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2x3=6个水平.

左边第1-4列分别为品种、年份、地点、区组代码

$$Y_{hijk} = \mu + \alpha_h + \beta_i + \gamma_j + \alpha\beta_{hi} + \alpha\gamma_{hj} + \beta\gamma_{ij} + \varepsilon_{hijk}$$

$h = 1, 2, 3; i, j, k = 1, 2$

1	1	1	1	19.7	1	1	1	1	28.6	1	1	1	1	20.3
1	1	1	2	31.4	1	1	1	2	38.3	1	1	1	2	27.5
1	1	2	1	40.8	1	1	2	1	44.4	1	1	2	1	44.6
1	1	2	2	29.4	1	1	2	2	34.9	1	1	2	2	41.4
1	2	1	1	45.5	1	2	1	1	47.5	1	2	1	1	54.2
1	2	1	2	50.3	1	2	1	2	41.1	1	2	1	2	52.3
1	2	2	1	53.9	1	2	2	1	63.7	1	2	2	1	53.9
1	2	2	2	58.8	1	2	2	2	61.1	1	2	2	2	59.1

思考题: 写出区组效应向量 e_B 及其系数矩阵 U_B

$$U_B = \begin{pmatrix} \beta_{1(1)} \\ \beta_{2(1)} \\ \beta_{1(2)} \\ \beta_{2(2)} \\ \beta_{1(3)} \\ \beta_{2(3)} \end{pmatrix}$$

左边第1-4列分别为品种、年份、地点、区组代码

$$Y_{hijk} = \mu + \alpha_h + \beta_i + \gamma_j + \alpha\beta_{hi} + \alpha\gamma_{hj} + \beta\gamma_{ij} + \varepsilon_{hijk}$$

$h = 1, 2, 3; i, j, k = 1, 2$

1	1	1	1	19.7	1	1	1	1	28.6	1	1	1	1	20.3
1	1	1	2	31.4	1	1	1	2	38.3	1	1	1	2	27.5
1	1	2	1	40.8	1	1	2	1	44.4	1	1	2	1	44.6
1	1	2	2	29.4	1	1	2	2	34.9	1	1	2	2	41.4
1	2	1	1	45.5	1	2	1	1	47.5	1	2	1	1	54.2
1	2	1	2	50.3	1	2	1	2	41.1	1	2	1	2	52.3
1	2	2	1	53.9	1	2	2	1	63.7	1	2	2	1	53.9
1	2	2	2	58.8	1	2	2	2	61.1	1	2	2	2	59.1

思考题: 写出区组效应向量 e_B 及其系数矩阵 U_B

$$y = Xb + U_Y e_Y + U_L e_L + U_{GY} e_{GY} + U_{GL} e_{GL} + U_B e_B + I e_e$$

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Why employ the mixed linear model (II)

$$y = Xb + \sum_{u=1}^r U_u e_u + e_{r+1} = Xb + U_1 e_1 + \dots + U_r e_r + e_{r+1}$$

其中 $V = \sum_{u=1}^r U_u V_u U_u^T + \sum_{u=r+1}^r (U_u C_{uu} U_u^T + U_r C_{uu} U_u^T) + \sigma_{r+1}^2 I$

随机效应分布假设 $e_u \sim MVN(0, V_u), u = 1, \dots, r; e_{r+1} \sim MVN(0, \sigma_{r+1}^2 I)$
 $cov(e_u, e_v) = C_{uv}, \text{ when } u \neq v$

- Include some fixed effects and some random effects
- Correlations between or within factors are allowed
- Can unbiasedly estimate parameters and their variances, and predict values of random effects
- Effectively analyze all kinds of complex genetic model or unbalanced data

思考题: 对真实生物学/医学等问题, 表型向量y是否一定符合多元正态分布?

不一定符合MVN. 因为可能有某些分量不满足MVN.

Genetic model that can not be analyzed by the ANOVA

- The effects between factors are mutually confound (Bio-model)
- coefficients of effects are not integers
- The effects between factors are correlated

Example: $y_{ijk} = \mu + A_i + A_j + \frac{1}{4}D_{ii} + \frac{1}{4}D_{jj} + \frac{1}{2}D_{ij} + e_{ijk}$

Example: $y'_{ijk} = \mu + A_i + A_j + \frac{1}{4}D_{ii} + \frac{1}{4}D_{jj} + \frac{1}{2}D_{ij} + C_i + Am_i + Am_j + Dm_{ij} + e_{ijk}$

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10.1 Mixed Linear Model

Matrix form for the regression model, ANOVA model (fixed model)

$$y = Xb + e$$

where b is a vector of fixed effects, e is a vector of random effects, with

$$e \sim MVN(0, \sigma_e^2 I)$$

So the dependent variable is also an independent normally distributed random vector,

$$y \sim MVN(Xb, \sigma_e^2 I)$$

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Mixed Linear Model (混合线性模型):

$$y = Xb + e = Xb + \sum_{u=1}^{r+1} U_u e_u$$

The vector of random effects e has a multivariate normal distribution

$$e \sim MVN(0, V)$$

The dependent variable is also a normally distributed random vector

$$y \sim MVN(Xb, V)$$

where V is a symmetric positive definite matrix.

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If there are r factors in the experimental design whose effects are random and independent, so there are $(r+1)$ random effects totally including the residual effects. Then the vector of random effect e can be composed into $(r+1)$ items

where $y = Xb + U_1 e_1 + U_2 e_2 + \dots + U_r e_r + e_{r+1}$

$$= Xb + \sum_u U_u e_u + I e_{r+1}$$

$$= Xb + (U_1 \ U_2 \ \dots \ I) e = Xb + Ue$$

U_u is the coefficient matrix of the u -th random factor

$$e_u \sim MVN(0, \sigma_u^2 I) \quad e_{r+1} \sim MVN(0, \sigma_{r+1}^2 I)$$

All the random factors are independent of each other

$$cov(e_u, e_v^T) = 0 \quad cov(e_u, e_{r+1}^T) = 0$$

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必须一个紧的一个松的.

$cov(e_u, e_v^T) = cov\left(\begin{pmatrix} e_{u1} \\ \vdots \\ e_{ui} \\ \vdots \\ e_{uq_u} \end{pmatrix}, (e_{v1} \ \dots \ e_{vj} \ \dots \ e_{vq_v})\right)$

$$= \begin{pmatrix} cov(e_{u1}, e_{v1}) & \dots & cov(e_{u1}, e_{vj}) & \dots & cov(e_{u1}, e_{vq_v}) \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ cov(e_{ui}, e_{v1}) & \dots & cov(e_{ui}, e_{vj}) & \dots & cov(e_{ui}, e_{vq_v}) \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ cov(e_{uq_u}, e_{v1}) & \dots & cov(e_{uq_u}, e_{vj}) & \dots & cov(e_{uq_u}, e_{vq_v}) \end{pmatrix}$$

$$= (cov(e_{ui}, e_{vj}))$$

q_u : 是向量 e_u 的大小, 即该效应的水平数
 q_v : 是向量 e_v 的大小, 即该效应的水平数

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The expectation and variance of the dependent variable

$$y = Xb + \sum_{u=1}^r U_u e_u + e_{r+1}$$

$$E(y) = E(Xb) + E\left(\sum_{u=1}^r U_u e_u + e_{r+1}\right)$$

$$= Xb + \sum_{u=1}^r U_u E(e_u) + E(e_{r+1})$$

$$= Xb$$

$$Var(y) = var(Xb) + var\left(\sum_{u=1}^r U_u e_u + e_{r+1}\right)$$

$$= \sum_{u=1}^r U_u var(e_u) U_u^T + var(e_{r+1})$$

$$= \sum_{u=1}^r U_u \sigma_u^2 U_u^T + \sigma_{r+1}^2 I$$

$$= \sum_{u=1}^r \sigma_u^2 U_u U_u^T + \sigma_{r+1}^2 I = V$$

$e_u \sim MVN(0, \sigma_u^2 I)$
 $e_{r+1} \sim MVN(0, \sigma_{r+1}^2 I)$
 $cov(e_u, e_v^T) = 0$

基于以上分布假设, 很容易推导左边等式

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MLM没有自由度, 不是基于F-test的.

The dependent variable is a normally distributed random vector

$$y \sim MVN(Xb, V = \sum_{u=1}^r \sigma_u^2 U_u U_u^T + \sigma_{r+1}^2 I)$$

Statistical analysis of mixed linear model:

- (1) Estimating each variance components σ_u^2
- (2) Predicting each random effects e_u
- (3) Estimating the fixed effects b

The analysis of mixed linear model can be used to perform unbiased analysis of unbalanced data or complex linear model.

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非平衡实验资料分析实例 (双列杂交),

	Male 1	Male 2	Male 3
Female 1	P1 (Missing)	P1×P2: 28.9, 12.63	P1×P3: 30.93, 14.98
Female 2	P2×P1: 28.38, 13.02	P2 (Missing)	P2×P3: 21.6, 11.4
Female 3	P3×P1: 27.31, 13.22	P3×P2 (Missing)	P3: 27.23, 12.89

- 试验共3个纯合亲本, 编号为: 1、2、3
- 母本、父本各提供一份配子给子代, 子代的性状变异受母本、父本基因的影响, 亲本 i 的基因效应记为 A_i
- 假设亲本 i 与亲本 j 交配产生 F_1 , 则 F_1 的遗传效应为 $A_i + A_j + D_{ij}$, 其中 D_{ij} 为两者间的互作效应 (显性效应, 杂种优势来源)

$$y_{ijk} = \mu + A_i + A_j + D_{ij} + \varepsilon_{ijk}$$

y_{ijk} 是亲本 i 与亲本 j 交配后代第 k 个个体的表型观测值

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非平衡实验资料分析实例 (双列杂交),

	Male 1	Male 2	Male 3
Female 1	P1 (Missing)	P1×P2: 28.9, 12.63	P1×P3: 30.93, 14.98
Female 2	P2×P1: 28.38, 13.02	P2 (Missing)	P2×P3: 21.6, 11.4
Female 3	P3×P1: 27.31, 13.22	P3×P2 (Missing)	P3: 27.23, 12.89

Female	Male	Replication	Y1	Y2
1	2	1	28.9	...
1	2	2	12.63	...
1	3	1	30.93	...
1	3	2	14.98	...
2	1	1	28.38	...
2	1	2	13.02	...
2	3	1	21.6	...
2	3	2	11.4	...
3	1	1	27.31	...
3	1	2	13.22	...
3	3	1	27.23	...
3	3	2	12.89	...

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	Male 1	Male 2	Male 3
Female 1	P1 (Missing)	P1×P2: 28.9, 12.63	P1×P3: 30.93, 14.98
Female 2	P2×P1: 28.38, 13.02	P2 (Missing)	P2×P3: 21.6, 11.4
Female 3	P3×P1: 27.31, 13.22	P3×P2 (Missing)	P3: 27.23, 12.89

$$y_{ijk} = \mu + A_i + A_j + D_{ij} + \varepsilon_{ijk}$$

假设不存在正反交差异, 即 $D_{ij} = D_{ji}$

$$y = 1\mu + U_A e_A + U_D e_D + I e_\varepsilon$$

$$\sim MVN(Xb, V = \sum_{u=1}^3 \sigma_u^2 U_u U_u^T)$$

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$$V = \sigma_A^2 U_A U_A^T + \sigma_D^2 U_D U_D^T + \sigma_\varepsilon^2 I$$

$$V = \sigma_A^2 \begin{pmatrix} 2 & 1 & 2 & 1 & 1 & 0 \\ 1 & 2 & 1 & 1 & 2 & 2 \\ 2 & 1 & 2 & 1 & 1 & 0 \\ 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 2 & 1 & 1 & 2 & 2 \\ 0 & 2 & 0 & 2 & 2 & 4 \end{pmatrix} + \sigma_D^2 \begin{pmatrix} 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix} + \sigma_\varepsilon^2 I$$

$$Q = V^{-1} - V^{-1} X (X^T V^{-1} X)^{-1} X^T V^{-1}$$

Prior values of variances $\sigma_A^2 = \sigma_D^2 = \sigma_\varepsilon^2 = 1$

将方差先验值代入上式V, 再计算Q, 记作 $Q_{(1)}$

$$Q_{(1)} = \begin{pmatrix} .5663 & -.0414 & -.4337 & -.0691 & -.0414 & .0193 \\ -.0414 & .5884 & -.0414 & -.0193 & -.4116 & -.0746 \\ -.4337 & -.0414 & .5663 & -.0691 & -.0414 & .0193 \\ -.0691 & -.0193 & -.0691 & .3011 & -.0193 & -.1243 \\ -.0414 & -.4116 & -.0414 & -.0193 & .5888 & -.0746 \\ .0193 & -.0746 & .0193 & -.1243 & -.0746 & .2348 \end{pmatrix}$$

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MINQUE(1) Estimation for Variances and Covariances:

$$[tr(U_u^T Q_{(1)} U_v U_v^T Q_{(1)} U_u)] [\hat{\sigma}_u^2] = [y^T Q_{(1)} U_u U_u^T Q_{(1)} y]$$

$$\begin{pmatrix} .5359 & .2707 & .1934 \\ .2707 & .5156 & .3683 \\ .1934 & .3683 & 2.2836 \end{pmatrix} \begin{pmatrix} \hat{\sigma}_A^2 \\ \hat{\sigma}_D^2 \\ \hat{\sigma}_e^2 \end{pmatrix} = \begin{pmatrix} y^T Q_{(1)} U_A U_A^T Q_{(1)} y \\ y^T Q_{(1)} U_D U_D^T Q_{(1)} y \\ y^T Q_{(1)} Q_{(1)} y \end{pmatrix}$$

Estimated variances For y_1 : $\hat{\sigma}_A^2 = 2.2555, \hat{\sigma}_D^2 = 6.2197, \hat{\sigma}_e^2 = 3.7666$
For y_2 : $\hat{\sigma}_A^2 = 0.5830, \hat{\sigma}_D^2 = 0.0511, \hat{\sigma}_e^2 = 0.8115$

$$[tr(U_u^T Q_{(1)} U_v U_v^T Q_{(1)} U_u)] [\hat{\sigma}_{uv}^2] = [y_1^T Q_{(1)} U_u U_u^T Q_{(1)} y_2]$$

$$\begin{pmatrix} .5359 & .2707 & .1934 \\ .2707 & .5156 & .3683 \\ .1934 & .3683 & 2.2836 \end{pmatrix} \begin{pmatrix} \hat{\sigma}_{A/A} \\ \hat{\sigma}_{D/D} \\ \hat{\sigma}_{e/e} \end{pmatrix} = \begin{pmatrix} y_1^T Q_{(1)} U_A U_A^T Q_{(1)} y_2 \\ y_1^T Q_{(1)} U_D U_D^T Q_{(1)} y_2 \\ y_1^T Q_{(1)} Q_{(1)} y_2 \end{pmatrix}$$

Estimated covariances $\hat{\sigma}_{A/A} = 1.5509, \hat{\sigma}_{D/D} = 0.4849, \hat{\sigma}_{e/e} = 1.6284$

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混合线性模型通式-1

$$y = X_1 b_1 + \dots + X_c b_c + U_1 e_1 + U_2 e_2 + \dots + U_r e_r + I e_{r+1}$$

$$= (X_1 \dots X_c) \begin{pmatrix} b_1 \\ \vdots \\ b_c \end{pmatrix} + (U_1 \dots U_r) \begin{pmatrix} e_1 \\ \vdots \\ e_r \end{pmatrix} + I e_{r+1} = Xb + \sum_{u=1}^r U_u e_u + e_{r+1}$$

$$Var(y) = V = \sum_{u=1}^r U_u V_u U_u^T + \sum_{u < v=1}^r (U_u C_{uv} U_v^T + U_v C_{vu} U_u^T) + \sigma_{r+1}^2 I$$

其中 随机效应分布假设 $e_u \sim MVN(0, V_u), u = 1, \dots, r$

$e_{r+1} \sim MVN(0, \sigma_{r+1}^2 I)$

$cov(e_u, e_v) = C_{uv}, \text{ when } u \neq v$

e_u 的方差协方差矩阵不一定是方阵, 说明其分量间可能相关, 但残差效应 e_{r+1} 的方差协方差矩阵为对角阵

不同随机效应间可能存在相关, 即存在 $cov(e_u, e_v) = C_{uv}$

这是一个非常非常通用的模型, 实际分析时一般不用

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混合线性模型通式-2

$$y = X_1 b_1 + \dots + X_c b_c + U_1 e_1 + U_2 e_2 + \dots + U_r e_r + I e_{r+1}$$

$$= (X_1 \dots X_c) \begin{pmatrix} b_1 \\ \vdots \\ b_c \end{pmatrix} + (U_1 \dots U_r) \begin{pmatrix} e_1 \\ \vdots \\ e_r \end{pmatrix} + I e_{r+1} = Xb + \sum_{u=1}^r U_u e_u + e_{r+1}$$

$$\sim MVN(Xb, V)$$

$$其中 V = \sum_{u=1}^r U_u \sigma_u^2 U_u^T + \sigma_{r+1}^2 I$$

随机效应分布假设 $e_u \sim MVN(0, \sigma_u^2 I), u = 1, \dots, r$
 $e_{r+1} \sim MVN(0, \sigma_{r+1}^2 I)$
 $cov(e_u, e_v) = 0, \text{ when } u \neq v$

对随机效应向量做了更强的分布假设
 e_u 各分量方差相同且相互独立
 e_u 与 $e_v (u \neq v)$ 间独立

随机效应也可能不具有正态分布, 但具有均值和方差结构, 即 $e_u \sim (0, V)$

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混合线性模型通式-2

$$y = X_1 b_1 + X_2 b_2 + \dots + X_c b_c + U_1 e_1 + U_2 e_2 + \dots + U_r e_r + I e_{r+1}$$

$$= Xb + \sum_{u=1}^r U_u e_u + e_{r+1} = Xb + Ue$$

e_i 的元素个数记为 $q_i, i = 1, 2, \dots, r+1$

$$= Xb + (U_1 \dots U_r I) \begin{pmatrix} e_1 \\ \vdots \\ e_r \\ e_{r+1} \end{pmatrix}$$

下标重新编码

$$= Xb + \sum_{k=1}^{c_1} u_k e_k + \sum_{k=c_1+1}^{c_2} u_k e_k + \dots + \sum_{k=c_{r-1}+1}^{c_r} u_k e_k + e_{r+1}$$

$$\sim MVN(Xb, V = \sum_{u=1}^r \sigma_u^2 U_u U_u^T + \sigma_{r+1}^2 I)$$

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混合线性模型通式-2

$$y = X_1 b_1 + X_2 b_2 + \dots + X_c b_c + U_1 e_1 + U_2 e_2 + \dots + U_r e_r + I e_{r+1}$$

$$= Xb + \sum_{u=1}^r U_u e_u + e_{r+1} = Xb + Ue$$

$$= Xb + \sum_{k=1}^{c_1} u_k e_k + \sum_{k=c_1+1}^{c_2} u_k e_k + \dots + \sum_{k=c_{r-1}+1}^{c_r} u_k e_k + e_{r+1}$$

$$\sim MVN(Xb, V = \sum_{u=1}^r \sigma_u^2 U_u U_u^T + \sigma_{r+1}^2 I)$$

其中 $Var(e_u) = \sigma_u^2 I, u = 1, 2, \dots, r+1$

$$U = [U_1 : U_2 : \dots : U_r : I] = [u_1 \dots u_{c_1} u_{c_1+1} \dots u_{c_2} u_{c_2+1} \dots u_{c_r} u_{c_r+1}]$$

$$e^T = [e_1^T : e_2^T : \dots : e_r^T : e_{r+1}^T]$$

$$= [e_1 \dots e_{c_1} e_{c_1+1} \dots e_{c_2} e_{c_2+1} \dots e_{c_r} e_{c_r+1}]$$

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$$y = X_1 b_1 + X_2 b_2 + \dots + X_c b_c + U_1 e_1 + U_2 e_2 + \dots + U_r e_r + I e_{r+1}$$

$$e_u \sim MVN(0, \sigma_u^2 I)$$

$$Cov(e_u, e_v) = 0, u = 1, \dots, r+1$$

$$Var(y) = V = \sum_{u=1}^r Var(U_u e_u) + Var(e_{r+1})$$

$$= U_1 \sigma_1^2 U_1^T + U_2 \sigma_2^2 U_2^T + \dots + U_r \sigma_r^2 U_r^T + \sigma_{r+1}^2 I$$

$$= \sigma_1^2 U_1 U_1^T + \sigma_2^2 U_2 U_2^T + \dots + \sigma_r^2 U_r U_r^T + \sigma_{r+1}^2 I$$

$$= \sigma_1^2 \sum_{k=1}^{c_1} u_k u_k^T + \sigma_2^2 \sum_{k=c_1+1}^{c_2} u_k u_k^T + \dots + \sigma_r^2 \sum_{k=c_{r-1}+1}^{c_r} u_k u_k^T + \sigma_{r+1}^2 I$$

$$U_1 U_1^T = (u_1 \dots u_{c_1}) \begin{pmatrix} u_1^T \\ \vdots \\ u_{c_1}^T \end{pmatrix} = \sum_{k=1}^{c_1} u_k u_k^T$$

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两个性状间的协方差及分解

$$y_1 = X_1 b_1 + X_2 b_2 + \dots + X_c b_c + U_1 e_1 + U_2 e_2 + \dots + U_r e_r + I e_{r+1}$$

$$y_2 = X_1 b_1 + X_2 b_2 + \dots + X_c b_c + U_1 e_1 + U_2 e_2 + \dots + U_r e_r + I e_{r+1}$$

y_1, y_2 分别样本性状1、性状2的表型观测值向量

$$\text{Cov}(e_i, e_j^T) = 0 \quad \text{Cov}(e_i, e_j^T) = 0 \quad \text{Cov}(e_i, e_j^T) = 0$$

$$\text{Cov}(e_i, e_j^T) = \sigma_{ij} \begin{pmatrix} 1 & & \\ & \ddots & \\ & & 1 \end{pmatrix} \quad i, j = 1, 2, \dots, r+1$$

$$\text{Cov}(y_1, y_2^T) = C$$

$$= \sigma_{1/1} U_1 U_1^T + \sigma_{2/2} U_2 U_2^T + \dots + \sigma_{r/r} U_r U_r^T + \sigma_{r+1/r+1} I$$

$$= \sigma_{1/1} \sum_{k=1}^{c_1} u_k u_k^T + \sigma_{2/2} \sum_{k=c_1+1}^{c_2} u_k u_k^T$$

$$+ \dots + \sigma_{r/r} \sum_{k=c_{r-1}+1}^{c_r} u_k u_k^T + \sigma_{r+1/r+1} I$$

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$$\text{Var}(y) = V = \sigma_1^2 U_1 U_1^T + \sigma_2^2 U_2 U_2^T + \dots + \sigma_r^2 U_r U_r^T + \sigma_{r+1}^2 I$$

$$= \sigma_1^2 \sum_{k=1}^{c_1} u_k u_k^T + \sigma_2^2 \sum_{k=c_1+1}^{c_2} u_k u_k^T$$

$$+ \dots + \sigma_r^2 \sum_{k=c_{r-1}+1}^{c_r} u_k u_k^T + \sigma_{r+1}^2 I$$

$$\text{Cov}(y_1, y_2) = C$$

$$= \sigma_{1/1} U_1 U_1^T + \sigma_{2/2} U_2 U_2^T + \dots + \sigma_{r/r} U_r U_r^T + \sigma_{r+1/r+1} I$$

$$= \sigma_{1/1} \sum_{k=1}^{c_1} u_k u_k^T + \sigma_{2/2} \sum_{k=c_1+1}^{c_2} u_k u_k^T$$

$$+ \dots + \sigma_{r/r} \sum_{k=c_{r-1}+1}^{c_r} u_k u_k^T + \sigma_{r+1/r+1} I$$

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10.2 The Analysis of Random Effects in Mixed Linear Model

Henderson method III

Charles Roy Henderson, 1911-1989

A giant in the field of animal breeding and genetics

M.S. degree in animal nutrition at Iowa State



https://www.researchgate.net/publication/n/13359838/Charles_Roy_Henderson_1911-1989_A_Brief_Biography

Scientific Contributions

In the fall of 1948, Henderson's professional career in Animal Breeding and Genetics began with his appointment as associate professor at Cornell University. That fall, he finished his thesis. His thesis had an important influence on the statistical computing package later developed by W. R. Harvey and also was a reference for several decades for graduate students at Iowa State. Nevertheless, material from his thesis does not seem to have been published. The reason, no doubt, is that shortly afterward he discovered a better method for which he was to become slowly but forever remembered: Henderson's mixed model equations (MME). The mixed model equations result from a simple modification of least squares equations. The simple modification results in best linear unbiased estimators (BLUE) of fixed effects and best linear unbiased predictors (BLUP) of random effects such as genetic values. Both BLUE and BLUP can be obtained in other far less computationally feasible ways. What Henderson did was transform a problem that was usually impossible into one that is often possible even with a large number of equations.

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10.2 The Analysis of Random Effects in Mixed Linear Model

Henderson method III

$$y = Xb + \sum_{u=1}^r U_u e_u + e_{r+1}$$

$$y = Xb + I e_{r+1}$$

SSR(b)

$$y = Xb + U_1 e_1 + I e_{r+1}$$

SSR(e₁|b)

$$y = Xb + U_1 e_1 + U_2 e_2 + I e_{r+1}$$

SSR(e₂|b, e₁)

$$y = Xb + U_1 e_1 + U_2 e_2 + U_3 e_3 + I e_{r+1}$$

SSR(e₃|b, e₁, ..., e_{r-1})

$$y = Xb + U_1 e_1 + U_2 e_2 + U_3 e_3 + \dots + U_r e_r + I e_{r+1}$$

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Henderson method III

$$y = Xb + \sum_{u=1}^r U_u e_u + e_{r+1} \sim MVN(Xb, V = \sum_{u=1}^r \sigma_u^2 U_u U_u^T + \sigma_{r+1}^2 I)$$

$$E(y^T A y) = \text{tr}(AV) + b^T X^T A X b$$

$$= \sum_{u=1}^r \sigma_u^2 \text{tr}(A U_u U_u^T) + \sigma_{r+1}^2 \text{tr}(A) + b^T X^T A X b$$

- 找到一系列的矩阵 A_i , 使其二阶式的期望 $E(y^T A_i y)$ 为随机效应方差的线性函数, 而与固定效应向量 b 无关, 即: $E(y^T A_i y) = \sum_{u=1}^r \sigma_u^2 \text{tr}(A_i U_u U_u^T) + \sigma_{r+1}^2 \text{tr}(A_i)$, $i = 1, 2, \dots, r+1$

- 从而得到如下一个线性方程组, 可以估计各方差, $y^T A_i y = \sum_{u=1}^r \sigma_u^2 \text{tr}(A_i U_u U_u^T) + \sigma_{r+1}^2 \text{tr}(A_i)$, $i = 1, 2, \dots, r+1$

显然, 可以取这样的 A_i , 使 $A_i X = 0$

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Henderson method III

$$y = Xb + \sum_{u=1}^r U_u e_u + e_{r+1} \sim MVN(Xb, V = \sum_{u=1}^r \sigma_u^2 U_u U_u^T + \sigma_{r+1}^2 I)$$

$$E(y^T A y) = \text{tr}(AV) + b^T X^T A X b$$

$$= \sum_{u=1}^r \sigma_u^2 \text{tr}(A U_u U_u^T) + \sigma_{r+1}^2 \text{tr}(A) + b^T X^T A X b$$

$$P_0 = X(X^T X)^{-1} X^T \quad A_0 = P_0$$

$$W_v = [X \ U_1 \ U_2 \ \dots \ U_v] \quad A_v = P_v - P_{v-1} \quad v = 1, 2, \dots, r$$

$$P_v = W_v (W_v^T W_v)^{-1} W_v^T \quad A_{r+1} = I - P_r$$

可以证明,

$$A_v X = 0, v = 1, 2, \dots, r$$

$$A_{r+1} X = 0$$

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$y = Xb + \sum_{u=1}^r U_u e_u + e_{r+1}$

Henderson method III
ANOVA table of mixed model (Henderson method III)

Cause of Variation	Degrees of Freedom	Sum of Squares	Expected Sum of Squares
(e, b)	$tr(A_1)$	$y^T A_1 y$	$\sum_{u=1}^r \sigma_u^2 tr(U_u^T A_1 U_u) + \sigma_{r+1}^2 tr(A_1)$
(e, b, e)	$tr(A_2)$	$y^T A_2 y$	$\sum_{u=1}^r \sigma_u^2 tr(U_u^T A_2 U_u) + \sigma_{r+1}^2 tr(A_2)$
...	...	...	...
(e, b, e, ..., e, e)	$tr(A_r)$	$y^T A_r y$	$\sigma_{r+1}^2 tr(A_r)$
Residual Effect	$tr(A_{r+1})$	$y^T A_{r+1} y$	$\sigma_{r+1}^2 tr(A_{r+1})$

Suppose the sums of squares are equal to their expected values, then we can obtain the estimator of each variance by solving the equation set.

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Discussion on the Henderson method III

Moment estimation method

Advantages:

- 不需要矩阵求逆就能估计各项随机效应方差分量
- 能分析非平衡数据
- 能分析混合线性模型。

Disadvantages:

- 不同随机效应向量 $e_1, e_2, \dots, e_r$ 加入到模型的顺序不同, 分析结果会不同
- 不能分析复杂模型。

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Maximum Likelihood (ML method)

$y = Xb + \sum_{u=1}^r U_u e_u + e_{r+1}$

Parameters: $\sigma_1^2, \dots, \sigma_{r+1}^2, b$

Assumption:

- $e_u \sim MVN(0, \sigma_u^2 I), u = 1, \dots, r$
- $e_{r+1} \sim MVN(0, \sigma_{r+1}^2 I)$
- $Cov(e_u, e_{r+1}) = 0$

Log likelihood function

$l = \log L = -\frac{1}{2} N \log(2\pi) - \frac{1}{2} \log |V| - \frac{1}{2} (y - Xb)^T V^{-1} (y - Xb)$

Maximizing

$\frac{\partial l}{\partial b} = X^T V^{-1} y - X^T V^{-1} Xb = 0$

$\frac{\partial l}{\partial \sigma_u^2} = -\frac{1}{2} tr(V^{-1} U_u U_u^T) + \frac{1}{2} (y - Xb)^T V^{-1} U_u U_u^T V^{-1} (y - Xb) = 0$

上面这 $r+1$ 个等式, 经改写后可得以下方程组

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Maximum Likelihood (ML method)

The ML equations for estimating the components of variance:

$[tr(U_u^T V^{-1} U_u U_u^T V^{-1} U_u)] [\hat{\sigma}_u^2] = [y^T Q U_u U_u^T Q y]$

where

$Q = V^{-1} - V^{-1} X (X^T V^{-1} X)^{-1} X^T V^{-1}$

$V = \sum_{u=1}^{r+1} \sigma_u^2 U_u U_u^T$

The ML estimation of the components of variance can be given by the following iteration.

$[tr(U_u^T V_{[h]}^{-1} U_u U_u^T V_{[h]}^{-1} U_u)] [\hat{\sigma}_{u[h+1]}^2] = [y^T Q_{[h]} U_u U_u^T Q_{[h]} y]$

where

$\hat{Q}_{[h]} = \hat{V}_{[h]}^{-1} - \hat{V}_{[h]}^{-1} X (X^T \hat{V}_{[h]}^{-1} X)^{-1} X^T \hat{V}_{[h]}^{-1}$

$\hat{V}_{[h]} = \sum_{u=1}^{r+1} \hat{\sigma}_{u[h]}^2 U_u U_u^T$

$\hat{\sigma}_{u[h]}^2$ is the estimate of σ_u^2 in the h -th iteration

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Discussion on ML method

$y = Xb + \sum_{u=1}^r U_u e_u + e_{r+1}$

Parameters: $\sigma_1^2, \dots, \sigma_{r+1}^2, b$

$l = \log L = -\frac{1}{2} N \log(2\pi) - \frac{1}{2} \log |V| - \frac{1}{2} (y - Xb)^T V^{-1} (y - Xb)$

- 其原理是基于似然函数最大求出参数的估计值, 那么固定效应 b 的估计是否会对方差的估计产生影响? 会受到影响, 没有无偏性, 是低估值。
- 方差分量必须是大于等于0的, 其解是否在解空间内?
- 方差估计的无偏性? 是有偏的, 低于真实的方差值。
- 观测值向量是否一定符合多元正态分布的假设?

不满足多元正态分布的假设。

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Maximum Likelihood (REML method)

$y = Xb + \sum_{u=1}^r U_u e_u + e_{r+1} \sim MVN(Xb, V = \sum_{u=1}^r \sigma_u^2 U_u U_u^T + \sigma_{r+1}^2 I)$

$K^T = (I - X(X^T X)^{-1} X^T) \Rightarrow K^T X = 0$

$y^* = K^T y = K^T (Xb + \sum_{u=1}^r U_u e_u + e_{r+1})$

$= 0b + K^T (\sum_{u=1}^r U_u e_u + e_{r+1}) = \sum_{u=1}^r K^T U_u e_u + K^T e_{r+1}$

$y^* = X^* b + \sum_{u=1}^r U_u^* e_u + e_{r+1}^* \sim MVN(0, V^* = K^T V K)$

新模型仍符合混合线性模型的矩阵向量形式, 因此仍可以利用 ML 等式来估计随机效应方差

与原模型相比, 固定/随机效应系数矩阵、表型向量、方差协方差矩阵出现了变化

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ML与REML $\sqrt{}$ 变为 Q 矩阵.

Example:

Maximum Likelihood (REML method)

$$y = Xb + \sum_{u=1}^r U_u e_u + e_{r+1} \sim MVN(Xb, V)$$

$$[tr(U_u^T V^{-1} U_u U_v^T V^{-1} U_v)] [\hat{\sigma}_u^2] = [y^T Q U_u U_u^T Q y] \quad u, v = 1, \dots, r+1$$

$$Q = V^{-1} - V^{-1} X (X^T V^{-1} X)^{-1} X^T V^{-1}$$

$$V = \sum_{u=1}^r \sigma_u^2 U_u U_u^T + \sigma_{r+1}^2 I$$

$$K^T y = K^T X b + \sum_{u=1}^r K^T U_u e_u + K^T e_{r+1}$$

$$y^* = X^* b + \sum_{u=1}^r U_u^* e_u + U_{r+1}^* e_{r+1} \sim MVN(0, V^* = K^T V K)$$

$$[tr(U_u^{*T} V^{*-1} U_u^* U_v^{*T} V^{*-1} U_v^*)] [\hat{\sigma}_u^2] = [y^{*T} Q^* U_u^* U_u^{*T} Q^* y^*]$$

$$Q^* = V^{*-1} - V^{*-1} X^* (X^{*T} V^{*-1} X^*)^{-1} X^{*T} V^{*-1}$$

$$V^* = \sum_{u=1}^r \sigma_u^2 U_u^* U_u^{*T} + \sigma_{r+1}^2 U_{r+1}^* U_{r+1}^{*T}$$

$$y^* = K^T y \quad X^* = K^T X = 0 \quad U_u^* = K^T U_u \quad V^* = K^T V K$$

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Restricted Maximum Likelihood (REML method)

The REML equations for estimating variance components:

$$[tr(U_u^T Q U_u U_v^T Q U_v)] [\hat{\sigma}_u^2] = [y^T Q U_u U_u^T Q y]$$

where $Q = V^{-1} - V^{-1} X (X^T V^{-1} X)^{-1} X^T V^{-1}$

$$V = \sum_{u=1}^{r+1} \sigma_u^2 U_u U_u^T$$

The REML estimation of the components of variance can be given by the following iteration,

$$[tr(U_u^T \hat{Q}_{[h]} U_v U_v^T \hat{Q}_{[h]} U_u)] [\hat{\sigma}_{u[h+1]}^2] = [y^T \hat{Q}_{[h]} U_u U_u^T \hat{Q}_{[h]} y]$$

where $\hat{Q}_{[h]} = \hat{V}_{[h]}^{-1} - \hat{V}_{[h]}^{-1} X (X^T \hat{V}_{[h]}^{-1} X)^{-1} X^T \hat{V}_{[h]}^{-1}$

$$\hat{V}_{[h]} = \sum_{u=1}^{r+1} \hat{\sigma}_{u[h]}^2 U_u U_u^T$$

$\hat{\sigma}_{u[h]}^2$ is the estimate of σ_u^2 in the h -th iteration

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Minimum Norm Quadratic Unbiased Estimation

The MINQUE estimator of the components of variance inferred by Rao

$$y = Xb + \sum_{u=1}^r U_u e_u + e_{r+1}$$

$$= Xb + Ue \sim MVN(Xb, V = \sum_{u=1}^r \sigma_u^2 U_u U_u^T + \sigma_{r+1}^2 I)$$

$$\begin{cases} e_u \sim MVN(0, \sigma_u^2 I) & e_{r+1} \sim MVN(0, \sigma_{r+1}^2 I) \\ Cov(e_u, e_v) = 0 & Cov(e_u, e_{r+1}) = 0 \end{cases}$$

MINQUE method doesn't need multivariate normal distribution assumptions

$$\begin{cases} e_u \sim (0, \sigma_u^2 I) & e_{r+1} \sim (0, \sigma_{r+1}^2 I) \\ Cov(e_u, e_v) = 0 & Cov(e_u, e_{r+1}) = 0 \end{cases}$$

Thus, $y = Xb + Ue \sim (Xb, V = \sum_{u=1}^r \sigma_u^2 U_u U_u^T + \sigma_{r+1}^2 I)$

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Minimum Norm Quadratic Unbiased Estimation

$$y = Xb + \sum_{u=1}^r U_u e_u + e_{r+1} = Xb + Ue \sim (Xb, V = \sum_{u=1}^r \sigma_u^2 U_u U_u^T + \sigma_{r+1}^2 I)$$

其中 $U = (U_1, U_2, \dots, U_r, I)$ $e = (e_1^T, e_2^T, \dots, e_r^T, e_{r+1}^T)^T = \begin{pmatrix} e_1 \\ \vdots \\ e_r \\ e_{r+1} \end{pmatrix}$

If each random effect e_u is known, then, $\hat{\sigma}_u^2 = \frac{1}{n_u} e_u^T e_u$, n_u is e_u 的大小

For any linear combination of σ_u^2 ($u = 1, \dots, r+1$),

$$\sum_{u=1}^{r+1} p_u \sigma_u^2 = p^T (\sigma_1^2, \dots, \sigma_{r+1}^2)^T = \sum_{u=1}^{r+1} \frac{p_u}{n_u} e_u^T e_u = e^T \Delta e$$

Could be estimated by a quadratic form

$$y^T A y = e^T U^T A U e \quad \text{with } A^T = A, \text{ and } A X = 0$$

Thus, minimizing the difference $e^T (U^T A U - \Delta) e$

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Minimum Norm Quadratic Unbiased Estimation

The MINQUE estimator of the components of variance

$$[tr(U_u^T Q_\alpha U_v U_v^T Q_\alpha U_u)] [\hat{\sigma}_u^2] = [y^T Q_\alpha U_u U_u^T Q_\alpha y]$$

where $Q_\alpha = V_\alpha^{-1} - V_\alpha^{-1} X (X^T V_\alpha^{-1} X)^{-1} X^T V_\alpha^{-1}$

$$V_\alpha = \sum_{u=1}^{r+1} \alpha_u^2 U_u U_u^T$$

α_u ($u = 1, 2, \dots, r+1$) is an artificially determined prior value.

Method of MINQUE estimation of the components of covariance between random vector y_1 and y_2 (Zhu, 1992):

$$[tr(U_u^T Q_\alpha U_v U_v^T Q_\alpha U_u)] [\hat{\sigma}_{u/v}] = [y_1^T Q_\alpha U_u U_u^T Q_\alpha y_2]$$

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MINQUE(1) supposes that all the prior values are 1 ($\alpha_u = 1$, $u = 1, 2, \dots, r+1$). The MINQUE(1) estimators can be provided by the following equations ($u, v = 1, 2, \dots, r+1$),

$$[tr(U_u^T Q_{(1)} U_v U_v^T Q_{(1)} U_u)] [\hat{\sigma}_u^2] = [y^T Q_{(1)} U_u U_u^T Q_{(1)} y]$$

where $Q_{(1)} = V_{(1)}^{-1} - V_{(1)}^{-1} X (X^T V_{(1)}^{-1} X)^{-1} X^T V_{(1)}^{-1}$

$$V_{(1)} = \sum_{u=1}^{r+1} U_u U_u^T$$

Estimator of covariance ($\hat{\sigma}_{u/v}$) in MINQUE(1):

$$[tr(U_u^T Q_{(1)} U_v U_v^T Q_{(1)} U_u)] [\hat{\sigma}_{u/v}] = [y_1^T Q_{(1)} U_u U_u^T Q_{(1)} y_2]$$

In MINQUE(0), the prior $\alpha_{r+1} = 1$, other prior values are 0

$$Q_{(0)} = I - X (X^T X)^{-1} X^T$$

$$V_{(0)} = I$$

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REML 中的 Q 变为 Q_α .

Best Linear Unbiased Prediction 随机效应预测.

Method of Unbiased Prediction of Random Effect

$$y = Xb + \sum_{u=1}^r U_u e_u + e_{r+1} \sim (Xb, V = \sum_{u=1}^r \sigma_u^2 U_u U_u^T + \sigma_{r+1}^2 I)$$

• **BLUP Methods**

$$\hat{e}_u = \sigma_u^2 U_u^T Q y \quad Q = V^{-1} - V^{-1} X (X^T V^{-1} X)^{-1} X^T V^{-1}$$

$$\hat{e}_u = \sigma_u^2 U_u^T \tilde{Q} y \quad (u = 1, \dots, r+1)$$

实际分析中仍需用样本数据来估计 σ_u^2 及 Q , 此时 $\hat{e}_u$ 已不是 y 的线性函数了

• **LUP Methods (Zhu & Weir, 1994, TAG, V89)**

$$\hat{e}_u = U_u^T Q_{(1)} y$$

• **AUP Methods (Zhu, 1993, Journal of Biomathematics, V8; Zhu & Weir, 1996, TAG, V92)**

$$\hat{e}_u = \kappa U_u^T Q_{(1)} y \quad \kappa = \sqrt{(n_u - 1) \hat{\sigma}_u^2 / (y^T Q_{(1)} U_u U_u^T Q_{(1)} y)}$$

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$$\text{Var}(\hat{e}) = \hat{\sigma}_u^2 I$$

Statistical testing on variances

For REML $[tr(U_u^T Q U_u U_v^T Q U_u)] [\hat{\sigma}_u^2] = [y^T Q U_u U_u^T Q y]$

$$Q = V^{-1} - V^{-1} X (X^T V^{-1} X)^{-1} X^T V^{-1}$$

$$V = \sum_{u=1}^{r+1} \sigma_u^2 U_u U_u^T$$

如果模型中的方差分量已知 (V 中的方差), 则

$$[\hat{\sigma}_u^2] = [tr(U_u^T Q U_u U_v^T Q U_u)]^{-1} [y^T Q U_u U_u^T Q y]$$

可以证明以下结论:

$$[\hat{\sigma}_u^2] \text{ asymptotically } MVN([\sigma_u^2], 2[tr(U_u^T Q U_u U_v^T Q U_u)]^{-1})$$

基于这一渐近分布, 可以构建统计量, 检验方差的显著性

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Statistical testing on variances

证明过程主要利用二阶式期望、方差的公式:

$$E(y^T A y) = tr(A \text{Var}(y)) + E(y^T) A E(y)$$

$$\text{Var}(y^T A y) = 2 tr((A \text{Var}(y))^2) + 4 E(y^T) A \text{Var}(y) A E(y)$$

$$E([\hat{\sigma}_u^2]) = [tr(U_u^T Q U_u U_v^T Q U_u)]^{-1} E(y^T Q U_u U_u^T Q y)$$

$$= [tr(U_u^T Q U_u U_v^T Q U_u)]^{-1} [tr(Q U_u U_u^T Q \text{Var}(y)) + E(y^T) Q U_u U_u^T Q E(y)]$$

上面右侧括号第二项为0, 因为:

$$Q E(y) = (V^{-1} - V^{-1} X (X^T V^{-1} X)^{-1} X^T V^{-1}) X b$$

$$= V^{-1/2} (I - P_{V^{-1/2} X} (X^T V^{-1/2} X)^{-1} X^T V^{-1/2}) V^{-1/2} X b$$

$$= V^{-1/2} (I - P_{V^{-1/2} X}) V^{-1/2} X b$$

$$= V^{-1/2} (V^{-1/2} X - P_{V^{-1/2} X} V^{-1/2} X) b$$

$$= V^{-1/2} (V^{-1/2} X - V^{-1/2} X) b = 0$$

$P_{V^{-1/2} X}$ 是矩阵 $(V^{-1/2} X)$ 的射影矩阵

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Statistical testing on variances

证明过程主要利用二阶式期望、方差的公式:

$$E(y^T A y) = tr(A \text{Var}(y)) + E(y^T) A E(y)$$

$$\text{Var}(y^T A y) = 2 tr((A \text{Var}(y))^2) + 4 E(y^T) A \text{Var}(y) A E(y)$$

$$E([\hat{\sigma}_u^2]) = [tr(U_u^T Q U_u U_v^T Q U_u)]^{-1} E(y^T Q U_u U_u^T Q y)$$

$$= [tr(U_u^T Q U_u U_v^T Q U_u)]^{-1} [tr(Q U_u U_u^T Q \text{Var}(y)) + E(y^T) Q U_u U_u^T Q E(y)]$$

$$= [tr(U_u^T Q U_u U_v^T Q U_u)]^{-1} [tr(Q U_u U_u^T Q \sum_{v=1}^{r+1} \sigma_v^2 U_v U_v^T)]$$

$$= [tr(U_u^T Q U_u U_v^T Q U_u)]^{-1} [tr(\sum_{v=1}^{r+1} Q U_u U_u^T Q \sigma_v^2 U_v U_v^T)]$$

$$= [tr(U_u^T Q U_u U_v^T Q U_u)]^{-1} [tr(\sum_{v=1}^{r+1} Q U_u U_u^T Q \sigma_v^2 U_v U_v^T)]$$

$$= [tr(U_u^T Q U_u U_v^T Q U_u)]^{-1} [tr(Q U_u U_u^T Q U_u U_u^T) \sigma_u^2 + \dots + tr(Q U_u U_u^T Q U_{r+1} U_{r+1}^T) \sigma_{r+1}^2]$$

$$= [tr(U_u^T Q U_u U_v^T Q U_u)]^{-1} [tr(Q U_u U_u^T Q U_u U_u^T)] [\sigma_u^2] = [\sigma_u^2]$$

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10.3 Estimating The Fixed Effect of Mixed Linear Model

For mixed linear model

$$y = Xb + \sum_{u=1}^r U_u e_u + e_{r+1}$$

$$\sim MVN(Xb, V = \sum_{u=1}^r \sigma_u^2 U_u U_u^T + \sigma_{r+1}^2 I)$$

$$e_u \sim MVN(0, \sigma_u^2 I); \text{Cov}(e_u, e_v) = 0, u \neq v$$

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Given all variance components, that is, the variance-covariance matrix V of dependent variable y is given, then the generalized least squares estimation (广义最小二乘估计法, GLS) can be used to estimate the parameter b of fixed effects in the model

$$\hat{b} = (X^T V^{-1} X)^{-1} X^T V^{-1} y$$

Since $\hat{b}$ is a product of the constant matrix

$A = (X^T V^{-1} X)^{-1} X^T V^{-1}$ and the normally distributed random vector y , thus $\hat{b}$ is also a normally distributed random vector.

$$\hat{b} \sim MVN_p(b, (X^T V^{-1} X)^{-1})$$

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有 V^{-1} 的就是 GLS.

BLUE 最佳线性无偏估计

GLS和OLS都是无偏估计

In practical application, we may perform linear unbiased estimation for the regression parameter b by ordinary least squares (OLS) estimation

$$\hat{b} = (X^T X)^{-1} X^T y$$

$\hat{b}$ is a random vector following multivariate normally distribution

$$\hat{b} \sim MVN_p(b, (X^T X)^{-1} X^T V X (X^T X)^{-1})$$

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V^{-1} 变为单位阵时就是OLS.

We may perform statistical test for the linear function of parameter ($c^T b$). For the null hypothesis $H_0: c^T b = m$ and its corresponding alternative hypothesis $H_1: c^T b \neq m$, if the statistic

$$|(c^T \hat{b} - c^T b) / \sqrt{c^T [(X^T X)^{-1} X^T V X (X^T X)^{-1}] c}| > z_{\alpha}$$

then the null hypothesis is rejected, while the alternative hypothesis is accepted.

0.05 $\rightarrow$ 196.

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10.4 Examples on the Application of Mixed Linear Model

QTL (数量性状基因座) 定位

$y = \mu + G + \varepsilon$

性状观测值 y 分解为群体均值 + 基因型效应 + 残差效应

QTL位点基因型为 QQ, Qq, qq , 其效应分别定义为 $a, d, -a$

a : 加性效应
 d : 显性效应

Figure 1 | Principles of mapping quantitative trait loci.

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Box 2 | Quantitative trait loci mapping methods

NATURE REVIEWS 2001

RIL群体各个体的表型(毛密度)

RIL群体各个体的基因型

多毛个体标记呈AA带型, 少毛个体呈aa带型, 中间型呈Aa带型

标记带型与毛密度有显著关系, 意味着标记附近可能存在控制毛的基因

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定位群体

$P_1 \times P_2$

产生配子: MQ, mQ, Mq, mq

F_1

Haploid $\rightarrow$ DH

BC_1 BC_2

F_2

连续自交 $\rightarrow$ RIL

F_1 IF_2

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Data of QTL mapping

Molecular marker code

MM=2, Mm=1, mm=0

M_ = 12, _m = 10

Individual	Molecular Marker						Value of Phenotype		
	M1	M2	M3	M4	M5	M6	Trait 1	Trait 2	Trait 3
Marker 1	1	1	1	12	0	0	9.1	10.4	38.73
2	0	0	1	12	2	2	10	10.4	38.67
3	2	2	0	0	0	0	10.7	10	38.64
Unknown	4	0	0	0	0	0	11	12.5	39.91
5	0	0	0	0	2	2	10.5	10.7	38.66
Marker 2	6	1	1	2	12	1	10.1	10.8	38.75
7	0	0	0	0	1	1	9.6	11.4	39.69
8	1	1	1	12	0	0	9.7	11	37.73
9	2	2	1	12	2	2	9.7	11.6	39.66
Marker 3	10	1	1	12	2	2	9.4	11.3	38.7
Unknown	11	2	2	1	12	1	8.5	10.6	38.28
? Gene 2	12	1	1	0	1	1	10	11.7	36.83
13	0	0	2	12	2	2	9.5	9.4	36.63
Marker 4	14	1	1	2	12	1	10	11.3	36.16

各个体标记基因型信息 各个体性状观测值

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10: Statistical Analysis of Mixed Linear Model

SCIENTIFIC REPORTS

OPEN **Mixed Linear Model Approaches of Association Mapping for Complex Traits Based on Omics Variants**

Received: 15 September 2018; Accepted: 05 April 2019; Published: 10 July 2019

Fu-Tao Zheng¹, Zhi-Hong Zhu², Xiao-Ran Yong³, Zhi-Ming Zhu⁴, Ting Qi⁵ & Jun Zhu⁶

Precise prediction for growth architecture of complex traits is impeded by the limited understanding on genetic effects of complex traits, especially on gene-by-gene (GxG) and gene-by-environment (GxE) interaction. In the past decades, an emphasis of high throughput technologies enables wide studies at multiple levels (such as genomics, transcriptomics, proteomics, and metabolomics). The analyses of large omics data, especially two-locus interaction analysis, are very time intensive. Integrating the diverse omics data and environmental effects in the analysis also remain challenges. We proposed mixed linear model approaches using GPU (Graphics Processing Unit) computation to simultaneously dissect various genetic effects. Analyses can be performed for estimating genetic main effects, GxG epistatic effects, and GxE environment interaction effects on large-scale omics data for complex traits, and for estimating heritability of specific genetic effects. Both mouse data on growth and human data on cholesterol demonstrated that genetic effects and environment interaction

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QTS定位和QTT/P/M定位的遗传模型

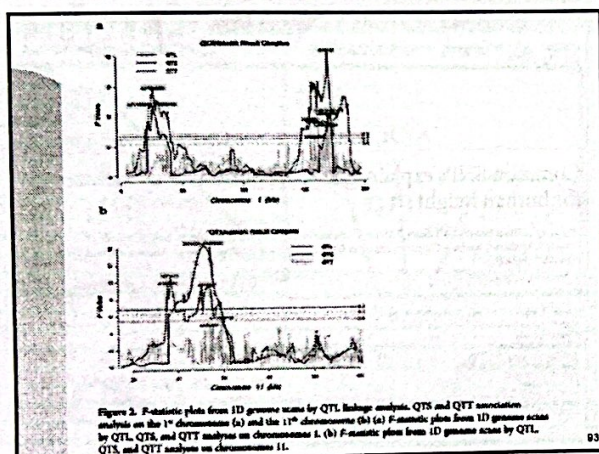
■ **QTS全模型:**

$$y_{ij} = \mu + e_j + \sum_k a_k u_{ik} + \sum_k d_k u_{ik} + \sum_{k < h} aa_{kh} u_{ik} u_{ih} + \sum_{k < h} ad_{kh} u_{ik} u_{ih} + \sum_{k < h} dd_{kh} u_{ik} u_{ih} + \sum_j \sum_k ae_{kj} u_{ik} + \sum_j \sum_k de_{kj} u_{ik} + \sum_j \sum_{k < h} aae_{khj} u_{ik} u_{ih} + \sum_j \sum_{k < h} ade_{khj} u_{ik} u_{ih} + \sum_j \sum_{k < h} dde_{khj} u_{ik} u_{ih} + \epsilon_{ij}$$

● **QTT/P/M全模型:**

$$y_{ij} = \mu + e_j + \sum_k q_k u_{ik} + \sum_{k < h} qq_{kh} u_{ik} u_{ih} + \sum_j \sum_k qe_{kj} u_{ik} + \sum_j \sum_{k < h} qqe_{khj} u_{ik} u_{ih} + \epsilon_{ij}$$

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课堂作业

$$y = Xb + \sum_{u=1}^r U_u e_u + e_{r+1} \sim MVN(Xb, V = \sum_{u=1}^r \sigma_u^2 U_u U_u^T + \sigma_{r+1}^2 I)$$

■ 对以上混合线性模型，为简化分析，常做哪些假设？

■ 模型中需要分析的参数有哪些？

■ 如何估计随机效应方差分量？估计公式？

■ 写出固定效应估计公式？

■ BLUP全称？写出BLUP预测公式

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① $e_u \sim MVN(0, \sigma_u^2 I)$ $e_{r+1} \sim MVN(0, \sigma_{r+1}^2 I)$
 $Cov(e_u, e_v) = 0$ $Cov(e_u, e_{r+1}) = 0$

② $\sigma_1^2, \dots, \sigma_{r+1}^2, b$

③ Henderson III, 最大似然法. Pr4, P38.

④ GLS, OLS.

⑤ Best Linear Unbiased Prediction

$$\hat{e}_u = \sigma_u^2 U_u^T Q y$$

$$\hat{e}_u = \sigma_u^2 U_u^T \hat{Q} y$$

$$Q = V^{-1} - V^{-1} X (X^T V^{-1} X)^{-1} X^T V^{-1}$$